Scaling of Large-Scale Motions in the Tropics

Purpose. We will use scale analysis to understand why large-scale tropical dynamics differ from familiar midlatitude quasi-geostrophic dynamics. The central questions are: Why are tropical pressure and temperature perturbations small? Why is dry vertical motion weak? Why does diabatic heating—especially latent heating—fundamentally change the tropical circulation?

Sources. These notes follow the scaling arguments in Charney (1963) and Roger Smith's lectures on Tropical Meteorology.

- Charney, J. G., 1963: A Note on Large-Scale Motions in the Tropics. *J. Atmos. Sci.*, **20**, 607–609.
- Roger Smith's Tropical Meteorology lectures.

Learning objectives. By the end of class, you should be able to:

1. explain why geostrophic scaling becomes less useful at low latitudes;
2. use the Rossby, Froude, and Richardson numbers to estimate tropical perturbation scales;
3. explain why dry synoptic-scale tropical motions are nearly horizontal and weakly coupled in the vertical;
4. show how diabatic heating changes the scale for vertical motion;
5. explain why strong tropical convection can coexist with small free-tropospheric temperature perturbations.

1. Context: What Was Different About the Mean Tropics?

From the previous lecture, recall several features of the mean tropical atmosphere. Complete the list with a phrase or short explanation.

Horizontal temperature gradients: ______________________________

Horizontal pressure gradients: ______________________________

Deep convection / latent heating: ______________________________

Large-scale overturning circulation: ______________________________

In-class exercise. Based only on these observations, which midlatitude approximation would you be most suspicious of applying unchanged in the deep tropics? Explain why.

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2. Starting Point: Hydrostatic Primitive Equations

For large-scale atmospheric motion, we begin with horizontal momentum, hydrostatic balance, mass continuity, and thermodynamics:

$$\left(\frac{\partial}{\partial t} + \boldsymbol{V}\cdot\boldsymbol{\nabla}_h\right)\boldsymbol{V} + w\frac{\partial \boldsymbol{V}}{\partial z} + f\boldsymbol{k}\times\boldsymbol{V} = -\frac{1}{\rho}\boldsymbol{\nabla}_h p, \tag{1}$$

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g, \tag{2}$$

$$\left(\frac{\partial}{\partial t} + \boldsymbol{V}\cdot\boldsymbol{\nabla}_h\right)\rho + \rho\,\boldsymbol{\nabla}_h\cdot\boldsymbol{V} + \frac{\partial(\rho w)}{\partial z} = 0, \tag{3}$$

$$\left(\frac{\partial}{\partial t} + \boldsymbol{V}\cdot\boldsymbol{\nabla}_h\right)\ln\theta + w\frac{\partial \ln\theta}{\partial z} = \frac{Q}{c_p T}. \tag{4}$$

Here $\boldsymbol{V} = (u, v)$, w is vertical velocity, p is pressure, ρ is density, θ is potential temperature, f is the Coriolis parameter, and Q is diabatic heating per unit mass.

Scaling question:

How large should the different terms and perturbations be?

In-class exercise. For each equation above, circle or annotate the term that represents: (a) Coriolis acceleration, (b) horizontal pressure-gradient acceleration, (c) vertical stratification, and (d) diabatic heating.

Class notes / reminders about the governing equations

3. Three Nondimensional Numbers: Three Physical Questions

We use the characteristic scales

- U =horizontal wind speed
- L=horizontal length scale
- D=vertical depth
- ρ_o = scale of density
- f = scale of the Coriolis parameter (vertical component of planetary vorticity)
- pressure scale height H_p, where:

$$H_p = \frac{p_0}{g\rho_0},$$

Rossby number	How strongly does Earth's rotation constrain the horizontal flow?
Froude number	How strong is the flow compared with gravity and pressure adjustment?
Richardson number	How strongly does stable stratification resist vertical motion?

3.1 Rossby Number: How Important Is Rotation?

To answer this question consider the parce acceleration (inertial) and Coriolis acceleration terms of equation 1. Their ratio is given the Rossby number:

$$\boxed{\mathrm{Ro} = \frac{U}{fL}}.$$

It compares the characteristic inertial acceleration,

$$\frac{U^2}{L},$$

with the Coriolis acceleration,

$$fU.$$

Question. Does Earth's rotation strongly constrain the motion, or is the flow's own inertia comparably important?

If

$$\boxed{\mathrm{Ro} \ll 1},$$

then rotation strongly constrains the flow and, consequently, **geostrophic balance is often a useful first approximation**. If

$$\boxed{\mathrm{Ro} \sim 1},$$

inertia and Coriolis accelerations are comparable and ageostrophic motion becomes important.

The tropical connection follows directly from

$$f = 2\Omega \sin\phi.$$

As latitude ϕ approaches the equator, f becomes small. Therefore, for the same U and L,

$$\boxed{f \downarrow \quad \Longrightarrow \quad \mathrm{Ro} \uparrow .}$$

For example, with

$$U = 10 \text{ m s}^{-1}, \qquad L = 1000 \text{ km} = 10^6 \text{ m},$$

we obtain

$$\mathrm{Ro} \sim 0.1 \quad \text{if} \quad f = 10^{-4} \text{ s}^{-1},$$

but

$$\mathrm{Ro} \sim 1 \quad \text{if} \quad f = 10^{-5} \text{ s}^{-1}.$$

$$\boxed{\mathrm{Ro} \ll 1: \text{ rotation strongly controls the flow}}$$

$$\boxed{\mathrm{Ro} \sim 1: \text{ rotation and inertia are comparably important}}$$

In-class exercise. Take $U = 10 \text{ m s}^{-1}$ and $L = 1000$ km. Estimate Ro for (i) $f = 10^{-4} \text{ s}^{-1}$ and (ii) $f = 10^{-5} \text{ s}^{-1}$. What changes dynamically as you move equatorward?

Class note: Why should we be cautious about applying ordinary geostrophic balance very close to the equator?

3.2 Froude Number: How Strong Is the Flow Compared with Gravity?

Using the pressure scale height,

$$\boxed{\mathrm{Fr} = \frac{U}{\sqrt{gH_p}}}.$$

The quantity

$$\sqrt{gH_p}$$

is a characteristic speed associated with gravity and pressure adjustment in the atmosphere.

Simple question. How fast is the atmospheric flow compared with the characteristic speed associated with gravity and pressure adjustment?

For representative synoptic-scale tropical values,

$$U \sim 10 \text{ m s}^{-1}, \qquad H_p \sim 10^4 \text{ m}, \qquad g \sim 10 \text{ m s}^{-2},$$

we obtain

$$\sqrt{gH_p} \sim 300 \text{ m s}^{-1},$$

so

$$\boxed{\mathrm{Fr} \sim 0.03}, \qquad \boxed{\mathrm{Fr}^2 \sim 10^{-3}}.$$

Thus the atmospheric flow is slow compared with the gravity/pressure-adjustment speed.

$$\boxed{\mathrm{Fr} \ll 1: \text{ gravity/pressure adjustment is strong compared with the flow}}$$

This becomes important because the pressure and temperature perturbation scales will contain Fr^2. Since $\mathrm{Fr}^2 \ll 1$, large-scale tropical pressure and temperature perturbations can remain small even when the circulation itself is dynamically important.

Simple analogy: Imagine pushing gently on a very stiff spring. The spring adjusts quickly but does not deform very much. In the same spirit, gravity and pressure adjustment make it difficult to maintain large departures of pressure and temperature from the background tropical state. We will discuss this more at a later time in our course.

In-class exercise. For $U = 15\ \mathrm{m\,s^{-1}}$ and $H_p = 10$ km, estimate Fr. Is the wind fast or slow compared with the gravity/pressure-adjustment speed?

Class note: In one sentence, what does $\mathrm{Fr} \ll 1$ **tell us?**

3.3 Richardson Number: How Difficult Is Vertical Motion?

From an earlier dynamics class, recall the stability parameter:

$$S_p = -\frac{T}{\theta_o}\frac{\partial \theta_o}{\partial p} \tag{5}$$

It is related to the the buoyancy frequency, which measures how rapidly an air parcel would oscillate vertically after being displaced in a stably stratified atmosphere.

$$N^2 = \frac{g}{\theta_0}\frac{d\theta_0}{dz}.$$

For the large-scale scaling used here, define the bulk Richardson number as

$$\boxed{\mathrm{Ri} = \frac{N^2 H_p^2}{U^2} = \left(\frac{N H_p}{U}\right)^2}.$$

Simple question. How strongly does stable stratification resist vertical displacement compared with the strength of the atmospheric flow?

For representative tropical-tropospheric values,

$$N \sim 10^{-2}\ \mathrm{s^{-1}}, \qquad H_p \sim 10^4\ \mathrm{m}, \qquad U \sim 10\ \mathrm{m\,s^{-1}},$$

we obtain

$$\boxed{\mathrm{Ri} \sim 100}.$$

A large Richardson number means that stable stratification strongly resists vertical displacement:

$$\boxed{\mathrm{Ri} \gg 1: \text{ the atmosphere strongly resists dry vertical motion}}.$$

A parcel displaced upward in a stably stratified atmosphere acquires a buoyancy anomaly relative to its surroundings and experiences a restoring force.

Simple analogy: Imagine pushing a ball uphill. A steeper hill makes upward motion more difficult. Strong stable stratification plays a similar role for vertical motion in the atmosphere.

In-class exercise. Using $N = 10^{-2}\ \mathrm{s}^{-1}$, $H_p = 10$ km, and $U = 10\ \mathrm{m\,s}^{-1}$, verify that $\mathrm{Ri} \sim 100$. What does this imply physically?

3.4 Put the Three Numbers Together

Number	Definition	Simple question	Typical tropical implication
Rossby	$\mathrm{Ro} = \dfrac{U}{fL}$	How strongly does rotation constrain the flow?	$\mathrm{Ro} \sim 1$: inertia and rotation are comparable.
Froude	$\mathrm{Fr} = \dfrac{U}{\sqrt{gH_p}}$	How strong is the flow compared with gravity/pressure adjustment?	$\mathrm{Fr} \ll 1$: pressure and temperature perturbations tend to be small.
Richardson	$\mathrm{Ri} = \dfrac{N^2 H_p^2}{U^2}$	How strongly does stratification resist vertical motion?	$\mathrm{Ri} \gg 1$: dry vertical motion is strongly suppressed.

Record the three ideas in your own words:

Rossby: ______

Froude: ______

Richardson: ______

4. How Large Are Pressure and Temperature Perturbations?

We now use the Rossby and Froude numbers together. The Rossby number tells us how strongly rotation enters the horizontal momentum balance, while the small Froude number will help explain why the resulting pressure and temperature perturbations remain small.

Quasi-geostrophic scaling of the momentum equations gives us (for small Rossby number):

$$f\boldsymbol{k} \times \boldsymbol{V} \sim -\frac{1}{\rho}\boldsymbol{\nabla}_h p, \tag{6}$$

Let δp, $\delta\rho$, and $\delta\theta$ denote characteristic departures from a horizontally uniform basic state. This gives us

$$fU \sim \frac{1}{\rho_o}\frac{\delta p}{L}$$

or,

$$\delta p \sim \rho_0 f U L.$$

Interestingly, even when $R_o \sim 1$, we get the same scaling for δp. This is because, when $R_o \sim 1$, the Coriolis and inertial effects are on the same order. In this case $fU \sim U^2/L \sim \frac{1}{\rho_o}\frac{\delta p}{L}$.

Using the hydrostatic pressure scale height

$$H_p = \frac{p_0}{g\rho_0},$$

we obtain

$$\frac{\delta p}{p_0} \sim \frac{fUL}{gH_p}.$$

Since

$$\mathrm{Ro} = \frac{U}{fL} \qquad \Longrightarrow \qquad fL = \frac{U}{\mathrm{Ro}},$$

this becomes

$$\boxed{\frac{\delta p}{p_0} \sim \frac{\mathrm{Fr}^2}{\mathrm{Ro}}}.$$

Hydrostatic and thermodynamic scaling give comparable fractional perturbations for density and potential temperature:

$$\boxed{\frac{\delta \rho}{\rho_0} \sim \frac{\delta\theta}{\theta_0} \sim \frac{\mathrm{Fr}^2}{\mathrm{Ro}}}.$$

Why does this matter?

If $\mathrm{Fr}^2 \sim 10^{-3}$, then changing from $\mathrm{Ro} \sim 0.1$ to $\mathrm{Ro} \sim 1$ changes the expected fractional pressure and temperature perturbations by about one order of magnitude.

	Midlatitudes	Low latitudes
Typical f	10^{-4} s^{-1}	10^{-5} s^{-1}
Ro	~ 0.1	~ 1
Fr^2	$\sim 10^{-3}$	$\sim 10^{-3}$
$\delta p/p_0$	$\sim 10^{-2}$	$\sim 10^{-3}$
$\delta\theta/\theta_0$	$\sim 10^{-2}$	$\sim 10^{-3}$

Central result. For comparable synoptic-scale disturbances, tropical pressure and temperature perturbations are smaller than their midlatitude counterparts.

In-class exercise. Suppose $\theta_0 = 300$ K. Estimate the characteristic $\delta\theta$ implied by the table for (a) midlatitudes and (b) low latitudes. What does this suggest about using temperature anomalies to identify tropical disturbances?

5. Dry Tropical Vertical Motion Is Extremely Weak

Now set diabatic source (condensational heating, radiative cooling etc.) to zero (i.e. $Q = 0$) in equation 4:

$$\frac{\partial \theta}{\partial t} + (\boldsymbol{V} \cdot \boldsymbol{\nabla}) \ln \theta + w \frac{\partial \ln \theta}{\partial z} = 0$$

Note that since $1/t \sim U/L$, $\frac{\partial \theta}{\partial t} \sim (U/L)\delta\theta$. Also, $(\boldsymbol{V} \cdot \boldsymbol{\nabla}) \ln \theta \sim (U/L)\delta\theta$. In other words, the LHS of the equation above scales as $(U/L)\delta\theta$.

Tus, in the absence of any diabatic heating/cooling, the dominant thermodynamic scaling is

$$\frac{U}{L}\frac{\delta\theta}{\theta_0} \sim W \frac{1}{\theta_0}\frac{d\theta_0}{dz},$$

where W is the characteristic vertical velocity.

Using

$$\frac{\delta\theta}{\theta_0} \sim \frac{\mathrm{Fr}^2}{\mathrm{Ro}}, \qquad N^2 = \frac{g}{\theta_0}\frac{d\theta_0}{dz},$$

we obtain

$$\frac{U}{L}\frac{\mathrm{Fr}^2}{\mathrm{Ro}} \sim W \frac{N^2}{g}.$$

With

$$\mathrm{Fr}^2 = \frac{U^2}{gH_p}, \qquad \mathrm{Ri} = \frac{N^2 H_p^2}{U^2},$$

and $D \sim H_p$,

$$\boxed{W \sim \frac{UD}{L}\frac{1}{\mathrm{RoRi}}}.$$

For

In-class exercise. Calculate W for typical values of synoptic scales in the midlatitudes and the tropics

Interpretation: In the absence of diabatic effects, how do tropical and extratropical vertical motions compare?

Connect the three numbers. For tropical synoptic-scale flow, $\mathrm{Ro} \sim 1$ means rotation and inertia are both important, $\mathrm{Fr} \ll 1$ keeps pressure and temperature perturbations small, and $\mathrm{Ri} \gg 1$ makes large dry vertical displacements difficult. Together these lead to very weak dry vertical motion.

6. Radiative Cooling and Slow Large-Scale Subsidence

The tropical troposphere experiences net radiative cooling. A representative rate is

$$\frac{Q}{c_p} \sim -1.2 \text{ K day}^{-1}.$$

Now, since we have already seen that the horizontal temperature gradients in the tropics are small. Similarly, the time change is also small. In this case, the thermodynamic equation 4 can be reduced to:

$$w\frac{\partial \ln \theta}{\partial z} \sim \frac{Q}{c_p T}$$

which gives:

$$\frac{w}{g}\left(\frac{g}{\theta}\frac{\partial \theta}{\partial z}\right) \sim \frac{Q}{c_p T}$$

which, neglecting perturbations in θ, switching $\theta \to \theta_o$, we haveL

$$\frac{w}{g}\left(\frac{g}{\theta_o}\frac{\partial \theta_o}{\partial z}\right) \sim \frac{Q}{c_p T}$$

Which can be written in terms of the buoyancy frequency (N^2) as:

$$w\frac{N^2}{g} \approx \frac{Q}{c_p T},$$

so

$$\boxed{w \approx \frac{g}{N^2}\frac{Q}{c_p T}}.$$

For

$$N = 10^{-2} \text{ s}^{-1}, \qquad T \sim 300 \text{ K}, \qquad \frac{Q}{c_p} \sim -1.2 \text{ K day}^{-1},$$

the scaling gives approximately

$$\boxed{w \sim -0.5 \text{ cm s}^{-1}}.$$

The negative sign denotes subsidence.

Class notes / interpretation

7. Moist Convection Changes the Scaling

The previous arguments assumed either $Q = 0$ or weak radiative forcing. In a precipitating disturbance, latent heating can be much larger. Then the thermodynamic balance becomes

$$\boxed{w\frac{N^2}{g} \approx \frac{Q}{c_p T}}.$$

Strong positive heating therefore supports upward motion:

$$Q > 0 \quad \Rightarrow \quad w > 0.$$

A representative latent-heating rate in tropical weather systems is

$$\frac{Q}{c_p} \sim 4.5 \text{ K day}^{-1},$$

which gives roughly

$$\boxed{w \sim 1.5 \text{ cm s}^{-1}}.$$

This is more than an order of magnitude larger than the dry adiabatic estimate. Effective static stability can also be smaller in convecting regions, permitting even larger vertical velocities.

Key contrast. Dry dynamics strongly suppress w, whereas latent heating permits substantial w and therefore strong vertical coupling.

In-class exercise. Using $w \propto Q/N^2$, predict what happens to the vertical-velocity scale if: (a) the heating rate doubles; (b) the effective static stability N^2 is cut in half.

Why can strong ascent occur without a large persistent temperature anomaly?

8. Heating, Rainfall, and Area Fraction

A rainfall rate of

$$1 \text{ cm day}^{-1}$$

corresponds to approximately

$$10 \text{ kg m}^{-2}\text{ day}^{-1}$$

of condensed water. With

$$L_v \approx 2.5 \times 10^6 \text{ J kg}^{-1},$$

the latent-energy release is approximately

$$2.5 \times 10^7 \text{ J m}^{-2}\text{ day}^{-1},$$

which corresponds to a deep-tropospheric mean heating rate of about

$$2.9 \text{ K day}^{-1}.$$

Thus a tropical-mean latent heating of about 0.9 K day^{-1} corresponds to roughly 0.3 cm day^{-1} of tropical-mean rainfall.

In-class exercise. If rain falls only over 20% of the tropical area but the area-mean rainfall is 0.3 cm day^{-1}, estimate the average rainfall rate within the raining region. Assume the other 80% receives no rain.

Strong narrow ascent, weak broad descent

Mass continuity requires upward and downward mass fluxes to balance:

$$A_\uparrow w_\uparrow + A_\downarrow w_\downarrow \approx 0.$$

Taking magnitudes,

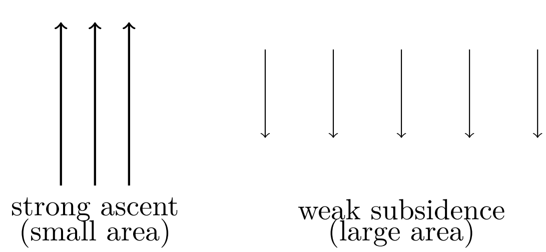
$$\boxed{\frac{A_\uparrow}{A_\downarrow} \sim \frac{|w_\downarrow|}{|w_\uparrow|}}.$$

For

$$w_\uparrow \sim 1.5 \text{ cm s}^{-1}, \qquad |w_\downarrow| \sim 0.5 \text{ cm s}^{-1},$$

we obtain

$$\frac{A_\uparrow}{A_\downarrow} \sim \frac{1}{3}.$$



In-class exercise. If $w_\uparrow = 2.5\ \mathrm{cm\,s^{-1}}$ and $|w_\downarrow| = 0.25\ \mathrm{cm\,s^{-1}}$, estimate $A_\uparrow/A_\downarrow$. What fraction of the combined area would be ascending?

C. One-minute summary

In 2–3 sentences, explain the central reason tropical large-scale dynamics cannot be understood simply by carrying midlatitude quasi-geostrophic intuition equatorward.

9. Questions to Carry Forward

1. If dry tropical disturbances have weak vertical coupling, what processes can organize disturbances through a deep tropospheric column?
2. How might moisture and convection alter the growth of tropical waves or vortices?
3. How does the weak-temperature-gradient character of the tropics affect the way we diagnose tropical weather compared with midlatitude weather?
4. Which parts of this scaling are likely to break down close to the equator, where $f \to 0$ and equatorial wave dynamics become important?

14. Free Class Notes